Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Artificial Intelligence (01CT0616)	**Aim:** To recommend an item to the user using collaborative-based filtering approach	
Experiment No: 12	**Date:**	**Enrolment No: 9230173022**

Aim: To recommend an item to the user using collaborative-based filtering approach

IDE: Google Colab

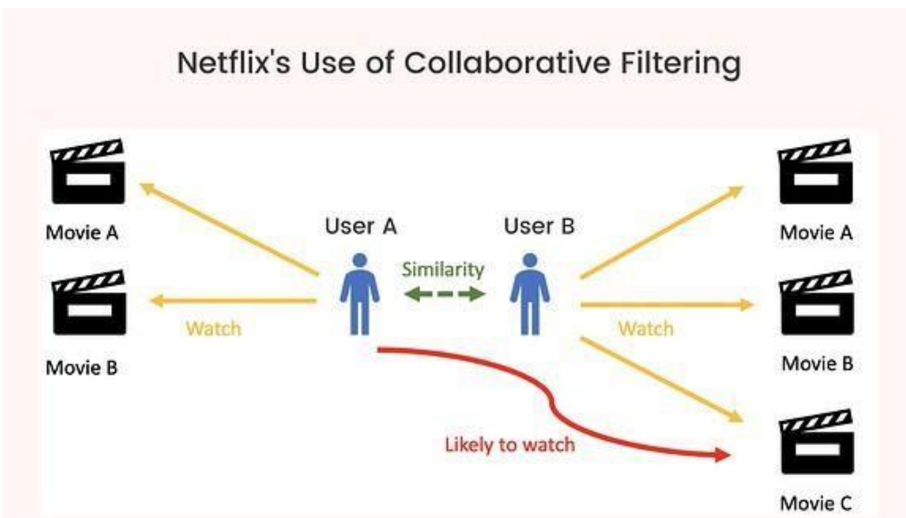
Theory:

There are a lot of applications where websites collect data from their users and use that data to predict the likes and dislikes of their users. This allows them to recommend the content that they like. Recommender systems are a way of suggesting or similar items and ideas to a user's specific way of thinking.

Recommender System is different types:

- **Collaborative Filtering:** Collaborative Filtering recommends items based on similarity measures between users and/or items. The basic assumption behind the algorithm is that users with similar interests have common preferences.
- **Content-Based Recommendation:** It is supervised machine learning used to induce a classifier to discriminate between interesting and uninteresting items for the user.

In Collaborative Filtering, we tend to find similar users and recommend what similar users like. In this type of recommendation system, we don't use the features of the item to recommend it, rather we classify the users into clusters of similar types and recommend each user according to the preference of its cluster.



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Pre Lab Exercise:

1. When can you use collaborative-based recommendation system approach?
   - Collaborative filtering is used when many users give ratings/reviews. It recommends items liked by similar users. Example: Netflix, Amazon, YouTube.
2. Write advantages of collaborative -based filtering approach
   - No need item features
   - Finds hidden user interests
   - Good recommendations with many users
   - Suggests unexpected items
3. Write disadvantages of collaborative -based filtering approach
   - Cold start problem
   - Sparse ratings issue
   - Needs many users/data
   - Slow for huge datasets

Program (Code):

```
!pip install opendatasets
import opendatasets as od

dataset_url = 'https://www.kaggle.com/datasets/luisreimberg/ratingscsv'
od.download(dataset_url)
import os
print(os.listdir('./ratingscsv'))
import pandas as pd
import numpy as np
from sklearn.metrics.pairwise import cosine_similarity
ratings = pd.read_csv('./ratingscsv/ratings.csv')
user_item = ratings.pivot_table(
   index='userId',
   columns='movieId',
   values='rating'
).fillna(0)
# Similarity in-between users
user_similarity = cosine_similarity(user_item)
user_similarity_df = pd.DataFrame(
   user_similarity,
   index=user_item.index,
   columns=user_item.index
)
user_similarity_df.head()
user_id = 10
recommend_movies(user_id, user_item, user_similarity_df)
```

```
user_item = ratings.pivot_table(
    index='userId',
    columns='movieId',
    values='rating'
).fillna(0)
# Similarity in-between users
user_similarity = cosine_similarity(user_item)
user_similarity_df = pd.DataFrame(
    user_similarity,
    index=user_item.index,
    columns=user_item.index
)
```

Results:

userId	1	2	3	4	5	6	7	8	9	10	...	601	602	603	604	605	606	607	608	609	610
userId																					
1	1.000000	0.027283	0.059720	0.194395	0.129080	0.128152	0.158744	0.136968	0.064263	0.016875	...	0.080554	0.164455	0.221486	0.070669	0.153625	0.164191	0.269389	0.291097	0.093572	0.145321
2	0.027283	1.000000	0.000000	0.003726	0.016614	0.025333	0.027585	0.027257	0.000000	0.067445	...	0.202671	0.016866	0.011997	0.000000	0.000000	0.028429	0.012948	0.046211	0.027565	0.102427
3	0.059720	0.000000	1.000000	0.002251	0.005020	0.003936	0.000000	0.004941	0.000000	0.000000	...	0.005048	0.004892	0.024992	0.000000	0.010694	0.012993	0.019247	0.021128	0.000000	0.032119
4	0.194395	0.003726	0.002251	1.000000	0.128659	0.088491	0.115120	0.062969	0.011361	0.031163	...	0.085938	0.128273	0.307973	0.052985	0.084584	0.200395	0.131746	0.149858	0.032198	0.107683
5	0.129080	0.016614	0.005020	0.128659	1.000000	0.300349	0.108342	0.429075	0.000000	0.030611	...	0.068048	0.418747	0.110148	0.258773	0.148758	0.106435	0.152866	0.135535	0.261232	0.060792

5 rows × 610 columns

	0
movieId	
78499	2.889794
69757	2.816518
72641	2.671646
318	2.648368
1721	2.558216

dtype: float64

Observation:

1. Learns hidden preferences.
2. Useful for movies/products/music apps.
3. Accuracy improves with large data.

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Post Lab Exercise:

1. Are you satisfied with the results/ratings shown by your designed model? If yes, justify. If no, describe the probable reason for under performance of the system.

➢ Yes, because recommendations are based on similar user preferences and rating patterns. If no, reason may be less data or sparse ratings.

2. Recommend top-10 "movies" using MovieLens 100K dataset (Link: https://grouplens.org/datasets/movielens/100k/). Attach the code with output. Also, write your observation for the same.

Top 10 movie recommendations for User ID 1:

movie_id	movie_title	predicted_rating
474	Dr. Strangelove or: How I Learned to Stop Worr...	4.001246
273	Heat (1995)	3.990894
433	Heathers (1989)	3.986334
566	Clear and Present Danger (1994)	3.594161
382	Adventures of Priscilla, Queen of the Desert, ...	3.405275
655	Stand by Me (1986)	3.398663
732	Dave (1993)	3.396734
684	In the Line of Fire (1993)	3.396734
403	Batman (1989)	3.368596